(1) $2Fe^{2+} + Tl^{3+} \rightarrow 2Fe^{3+} + Tl^{+}$

$$-\frac{d[Tl^{3+}]}{dt} = \frac{k\,[Fe^{2+}]^2\,[Tl^{3+}]}{[Fe^{2+}] + k'\,[Fe^{3+}]}$$

$$\rightleftharpoons \quad I + Fe^{3+}$$

$$I + Fe^{2+} \longrightarrow$$

Assume 2nd step is slow

$$v = \frac{k\,[Fe^{2+}]^2\,[Tl^{3+}]}{[Fe^{3+}]}$$

stoichiometry of RDS $= 2Fe + Tl - Fe$

$\rightarrow Fe + Tl$

Charge of RDS $= 4 + 3 - 3 = 4$

$I \rightarrow$ One Tl and charge +2

Tl^{2+}

$$Fe^{2+} + Tl^{3+} \rightleftharpoons Tl^{2+} + Fe^{3+}$$

$$Tl^{2+} + Fe^{2+} \longrightarrow Fe^{3+} + Tl^{+}$$

$$2Fe^{2+} + Tl^{3+} \longrightarrow 2Fe^{3+} + Tl^{+}$$

② $C(CH_3)_3Cl + OH^- \rightarrow C(CH_3)_3OH + Cl^-$

$$v = \frac{k[C(CH_3)_3Cl][OH^-]}{k'[Cl^-] + k''[OH^-]}$$

$$\rightleftharpoons \quad I + Cl^-$$

$$I + OH^- \longrightarrow$$

$k'' << k'$

$$v = k\frac{[C(CH_3)_3Cl][OH^-]}{k'[Cl^-]}$$

RDS $\quad C(CH_3)_3$

Charge $\Rightarrow 0$

$I \Rightarrow C(CH_3)_3^+$

$$C(CH_3)_3Cl \rightleftharpoons C(CH_3)_3^+ + Cl^-$$

$$C(CH_3)_3^+ + OH^- \rightarrow C(CH_3)_3OH$$

(3) $$Hg_2^{2+} + Tl^{3+} \rightarrow 2Hg^{2+} + Tl^{+}$$

$$\text{Rate} = k \frac{[Hg_2^{2+}][Tl^{3+}]}{[Hg^{2+}]}$$

$$\rightleftharpoons \quad I + Hg^{2+}$$

$$I \longrightarrow$$

Stoichiometry

$$2Hg + Tl - Hg \longrightarrow Hg + Tl$$

Charge $\quad 2+3-2 = 3$

$$Hg_2^{2+} \longrightarrow Hg + Hg^{2+}$$

$$Hg + Tl^{3+} \longrightarrow Hg^{2+} + Tl^{+}$$

$$Hg_2^{2+} + Tl^{3+} \longrightarrow Hg^{2+} + Tl^{+}$$

(5) $Ph_2CHCl + H_2O \rightarrow Ph_2CHOH + Cl^- + H^+$

$$\frac{d(Ph_2CHOH)}{dt} = \frac{\alpha[Ph_2CHCl]}{\beta + [Cl^-]}$$

$$\rightleftharpoons \quad Cl^- + I$$

$$I \longrightarrow P$$

Stoichiometry of RDS = Ph_2CH

Chage = $+1$

$$Ph_2CHCl \longrightarrow Ph_2CH^+ + Cl^-$$

$$Ph_2CH^+ + H_2O \longrightarrow Ph_2CHOH + H^+$$

④ Unimolecular Reaction

$$\text{Rate law} = \frac{k[A]^2}{k'[A]+k''}$$

High Concentration of [A]

First order reaction

Low Concentration of [A]

Second order reaction

$$\rightleftharpoons A + I$$

$$I \longrightarrow P$$

$k'' \ll k'$

stoichiometry → one A

$k' \ll k''$

stoichiometry → two A

$$A + A \longrightarrow A + A^*$$

$$A^* + A \xrightarrow{k''} A + A$$

$$I \xrightarrow{k'} P$$

stoichiometry of I is A